

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES

MONDAY, 20 MAY 2024

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in Section A and **ONLY ONE (1)** question in Section B. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are /are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES**Section A (Compulsory Questions)**

- Q1. (a) Explain any **FOUR (4)** necessities of planning in advance, in an incremental development approach. (8 marks)
- (b) Suggest the most appropriate generic software process model between “plan-driven” and “agile approach” that might be used as a basis for managing the development of the following systems. Explain your selection with justification.
- (i) An inventory management system for a large retail chain. (2 marks)
 - (ii) An e-learning platform for a university. (2 marks)
 - (iii) An embedded software controlling a smart thermostat in a smart home system. (2 marks)
 - (iv) A content management system (CMS) for a news website. (2 marks)
 - (v) A flight reservation system for an airline. (2 marks)
- (c) State the **SEVEN (7)** roles of scrum master. (7 marks)
[Total : 25 marks]
- Q2. (a) Imagine you are tasked with developing a simple online bookstore application using the MVC (Model-View-Controller) architecture. The application allows users to browse books, add the books to the shopping cart, and place orders. Based on the given scenario:
- (i) Illustrate MVC architecture for this online bookstore applications. (9 marks)
 - (ii) State the role of each component in the MVC architecture for this online bookstore application. (7 marks)
- (b) Discuss any **THREE (3)** advantages of using MVC architecture. Give relevant examples for each point. (9 marks)
[Total : 25 marks]

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Q3. (a) Component-based software engineering (CBSE) is an approach to software development that relies on the reuse of entities called “**Software components**”. Explain the below characteristics of components:

(i) Independent. (4 marks)

(ii) Standardized. (4 marks)

(b) As a software engineer of SEN IT Solution, you are required to develop Customer Order System. You have decided to use Component Based Software Engineering approach. The customer order system should have functions to facilitate the customers to manage their orders. Based on the given scenario:

(i) Identify any **TWO (2)** “provides” interface. (2 marks)

(ii) Identify any **FOUR (4)** “requires” interface. (4 marks)

(iii) Illustrate the model for data collector component using UML notations based on your answers in Q3(b)(i) and Q3(b)(ii). (4 marks)

[Hints: Combine the answer (i), (ii) and (iii) into a diagram]

(c) State **THREE (3)** drawbacks of adopting “**Waterfall**” model as software development methodology. (7 marks)

[Total : 25 marks]

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES**Section B (Choose any ONE Question)**

Q4. (a) As a software tester at Kanniapa IT Solutions, you need to test a system that handles user ages, with valid age criteria set between 18 and 60. Given the above scenario:

(i) Define all the criteria for partition testing for the system that processes user age. (3 marks)

(ii) Generate **FIVE (5)** test cases representing each scenario based on your answer in Q4(a)(i). (7 marks)

(b) Kannan, an academic coordinator at a university, uses the EduConnect system to manage resources and support faculty members. Kannan logs into the EduConnect system to access the academic schedule and course information. He reviews faculty schedules, classroom allocations, and course details for the upcoming semester. Kannan identifies specific course materials, lecture notes, and research papers needed for faculty development. He selects relevant resources to download onto his laptop for offline access during faculty meetings and workshop.

During a faculty development session, a professor, Dr. Patel, raises concerns about the relevance of certain course materials. Kannan decrypts Dr. Patel's records, reviews the course materials in question, and checks for updates or alternative resources. After the session, Kannan updates the EduConnect system with the reviewed course materials. The system automatically generates a call list for Kannan, reminding him to follow up with faculty members about the updated resources and gather feedback

Analyze the above scenario and determine any **FOUR (4)** features tested by the system flow. (8 marks)

(c) Process standards in quality managements defines the processes that should be followed during software development. In manufacturing, it is known that product quality is closely related to the production process. Illustrate a generic **process-based approach** diagram to achieving product quality. (7 marks)

[Total : 25 marks]

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- Q5. (a) Software systems are critical business assets for an organization. In order to maintain the value of these assets to the business, they must be changed and updated often. Most of the software budget in large companies is devoted to changing and evolving existing software rather than developing new software.
- (i) Specify the elements on which software evolution process depends. (4 marks)
 - (ii) Illustrate the software evolution process. (8 marks)
- (b) Legacy systems are older systems that rely on languages and technology that are no longer used for new systems development. Determine the legacy system component for each of the below scenario.
- (i) A manufacturing company, founded in the 1980s, has been using a production control system running on custom-built hardware since then. Recently, they faced a breakdown, and sourcing replacement parts became impossible. They are forced to halt their business operation. (2 marks)
 - (ii) A healthcare institution uses a patient records system developed in the '90s. The system is written in a language no longer in use. This makes the process of adapting to evolving medical record standards is challenging. (2 marks)
 - (iii) A regional bank has been relying on a core banking system since the early 2000s. The middleware connecting various banking modules, is no longer supported by the vendor. The bank experiences frequent disruptions. (2 marks)
 - (iv) An e-commerce company relies on an order processing system with data stored in multiple databases. Over time, data inconsistencies lead to shipping errors and customer dissatisfaction. (2 marks)
- (c) Configuration Management (CM) is an approach to systematically manage the policies, processes and tools for handling changing software systems. Explain any **TWO (2)** activities in configuration management. (5 marks)
- [Total : 25 marks]
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